

Week 7 · Laboratory Problems — build the measurement path in Simulink

- Course: Sensor Signal Processing and Fusion · Department of Autonomous Vehicle System Engineering
- Time: **one hour**, immediately after the Week 7 lecture hour
- Three problems, in order.

What this hour is for

Every week so far handed the controller a clean state. A real sensor reports what the vessel really does — including the part of it that is the sea.

This hour builds the path between the hull and the autopilot, and then asks the question that decides its shape: **which part of what is measured should be acted on?** Problem 1 removes what cannot be followed and measures what that saves. Problem 2 switches the sea off and measures what the same filter costs. Problem 3 adds a disturbance that is slow, shows that the filter passes it, and removes it with the one term Week 4 left out.

The autopilot's gains are **not touched** in any of the three. A sea is not answered by retuning.

Before starting

```
cd lectures/W07_simulink/problems
W07_P1_start           % creates W07_P1.slx — the sea, the hull, nothing between
W07_check(1)           % run this whenever, as often as needed
```

The solver is already set to fixed-step `ode4` at `h = 0.02` s. **Do not change it.**

★ One requirement on every model

flog — To Workspace, **Structure With Time**, carrying $[\psi_m; \psi_f; N]$: the measurement, **what the controller actually uses**, and the moment it asks for, in degrees, degrees and newton-metres.

ψ_f is logged because no outcome can say whether the filter is *in the loop*. A model can steer well with the notch built and never connected. With $\zeta_n = \zeta_d$ the checker requires $\psi_f = \psi_m$ **exactly**, and with the notch in it requires them to differ — which is a structural test, not a performance one.

The sea, the hull, the commanded heading, the slow disturbance and Week 6's allocation are all provided. The wave is one fixed realisation of a JONSWAP spectrum, so every student's sea is the same sea and the lecture's numbers are a fair pass mark.

⚠ The filter is switched off by a number, never by deleting a block

Setting $\zeta_n = \zeta_d$ makes $H(s) = 1$ identically. The same model then reproduces the unfiltered one **exactly**, and the comparison is between two runs of one model rather than between two models that might differ elsewhere. The checker relies on this and tests it.

Problem 1 · The measurement, and the notch (25 minutes)

Build. Three things, in this order.

$$\psi_m = \psi + \psi_w, \quad r_m = r + r_w$$

$$H(s) = \frac{s^2 + 2\zeta_n\omega_0 s + \omega_0^2}{s^2 + 2\zeta_d\omega_0 s + \omega_0^2}, \quad \zeta_n < \zeta_d, \quad |H(j\omega_0)| = \frac{\zeta_n}{\zeta_d}$$

$$N = K_p \text{ssa}(\psi_d - \psi_f) - K_d r_f, \quad |N| \leq N_{\max}$$

Symbol	Quantity	Value · source
ω_0	the peak frequency of the sea	$2\pi/T_0 = 3.14$ rad/s · §7-2
ζ_n, ζ_d	the depth and the width of the notch	0.05 and 0.30 · §7-4
K_p, K_d	the heading gains	300, 100 · Week 4, unchanged

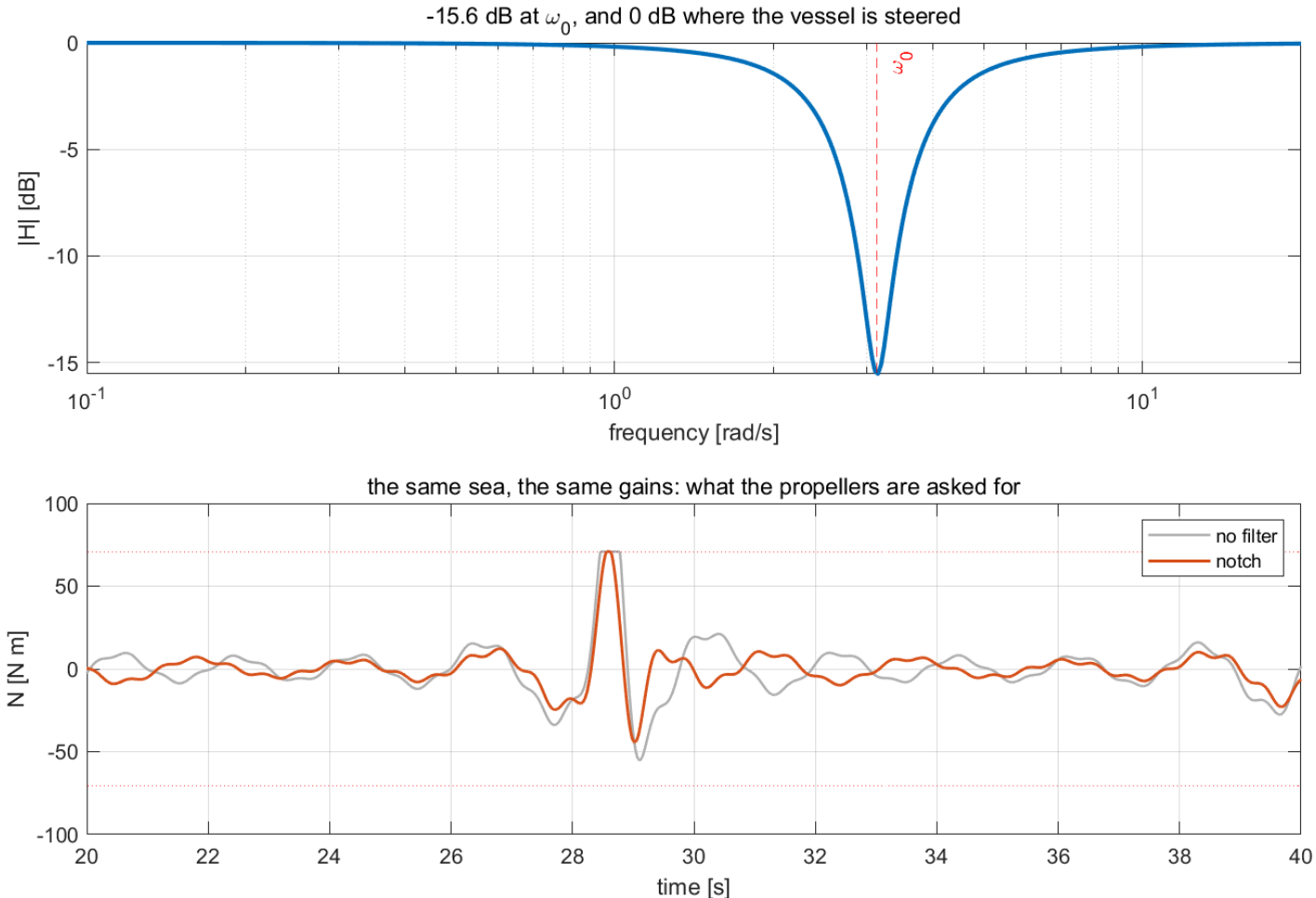
★ Filter the rate as well as the heading

The wave carries **11.64** deg/s of yaw rate, which is as much as this vessel can produce. A notch on the heading alone leaves most of the problem in the loop, through the D term.

Verify. `W07_check(1)` . The same sea is flown twice, once with $\zeta_n = \zeta_d$ and once with $\zeta_n = 0.05$.

What the checker expects	no filter	notch
heading std, $t > 20$ s	1.551°	1.042°
moment std, $t > 20$ s	18.85 N·m	13.41 N·m
ψ_f against ψ_m	identical, to 10^{-6}	visibly different

What a correct model produces



Reading the figure	
top, the dip	−15.6 dB at the marked ω_0 , which is $20\log_{10}(\zeta_n/\zeta_d)$
top, away from the dip	back to 0 dB on both sides: everything else is passed
bottom, grey against blue	the same sea and the same gains, a visibly quieter demand
bottom, the flat tops that remain	the notch attenuates by −15.6 dB; it does not delete
the check	if the heading std does not move, the filtered signals are built but not wired into the autopilot — look at the <code>what it filters</code> column

The point. One filter, two numbers, and **every column improves at once**. Nothing was retuned: the vessel steers better because it is no longer being shaken.

Problem 2 · What it costs (15 minutes)

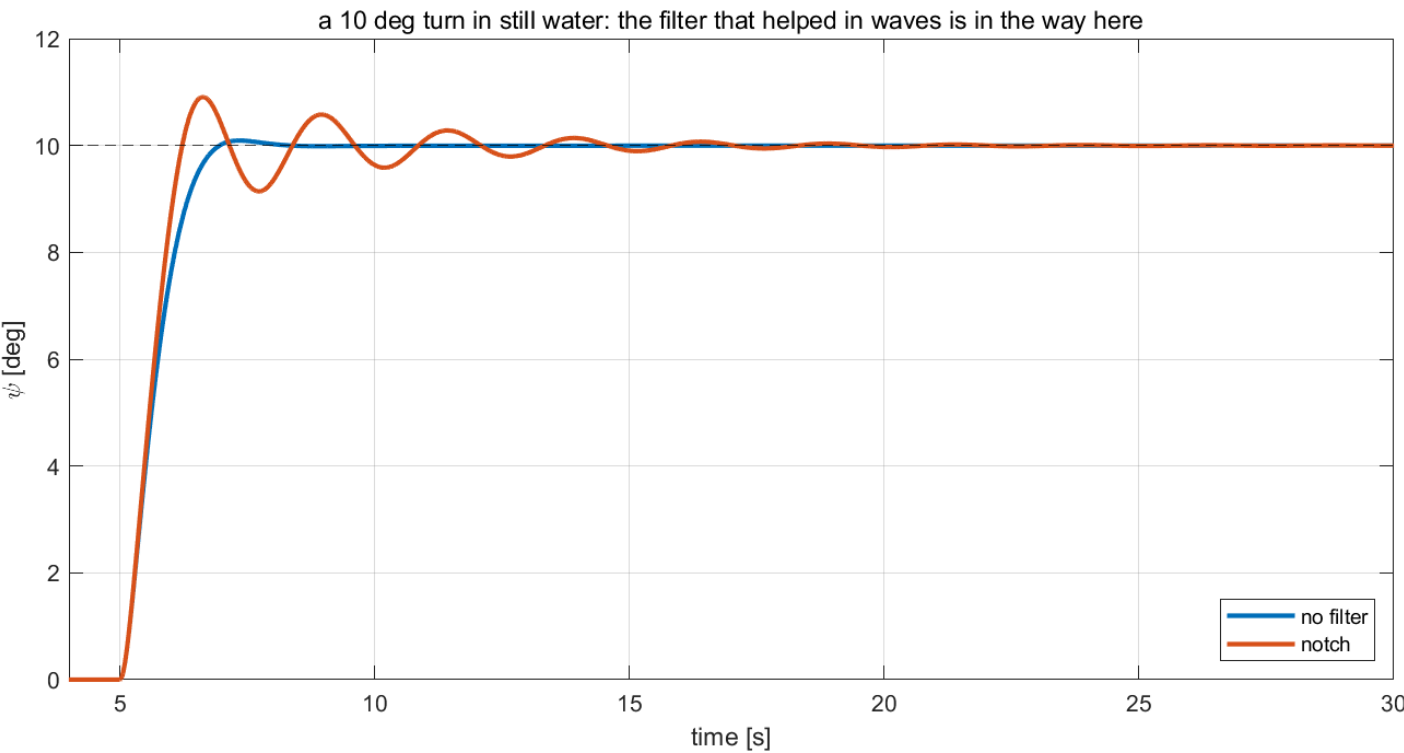
Build. Nothing. Set `wave_on = 0` and give the same 10° step, once with $\zeta_n = \zeta_d$ and once with the notch.

Predict before running. The notch adds phase where it attenuates, and some of that phase reaches the frequency at which the vessel is steered. Decide, before looking, whether that will show up as overshoot, as settling time, or as both.

Verify. `w07_check(2)` .

What the checker expects	no filter	notch
overshoot	0.93 %	9.07 %
settling	1.72 s	7.76 s
phase of H at 1.6 rad/s	0.0°	−18.5°

What a correct model produces



Reading the figure	
the two traces before the peak	almost the same, the notched one slightly faster
the peak of the notched trace	9.07 % against 0.93 %
the tail of the notched trace	a long slow approach, 7.76 s
the check	if the two traces coincide, <code>wave_on = 0</code> has switched off the filter rather than the sea

The point. The filter that improved every column of Problem 1 makes this manoeuvre visibly worse, and **both are true at once**. Attenuation is bought with phase, and phase is what damping is made of — the same trade Week 2 §2-10 paid for the derivative, paid again here.

Problem 3 · The slow part must not be filtered (20 minutes)

Build. One term, in the autopilot of Problem 1:

$$u = K_p e + I - K_d r_f, \quad N = \text{sat}(u), \quad I \leftarrow I + h(K_i e + K_b(N - u))$$

The back-calculation is Week 2 §2-11, unchanged.

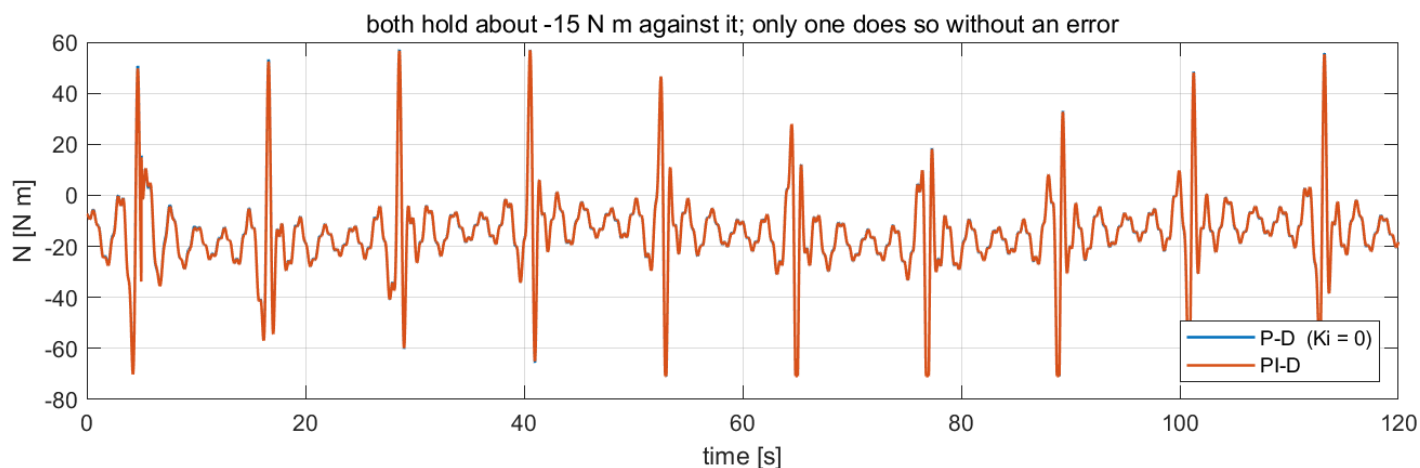
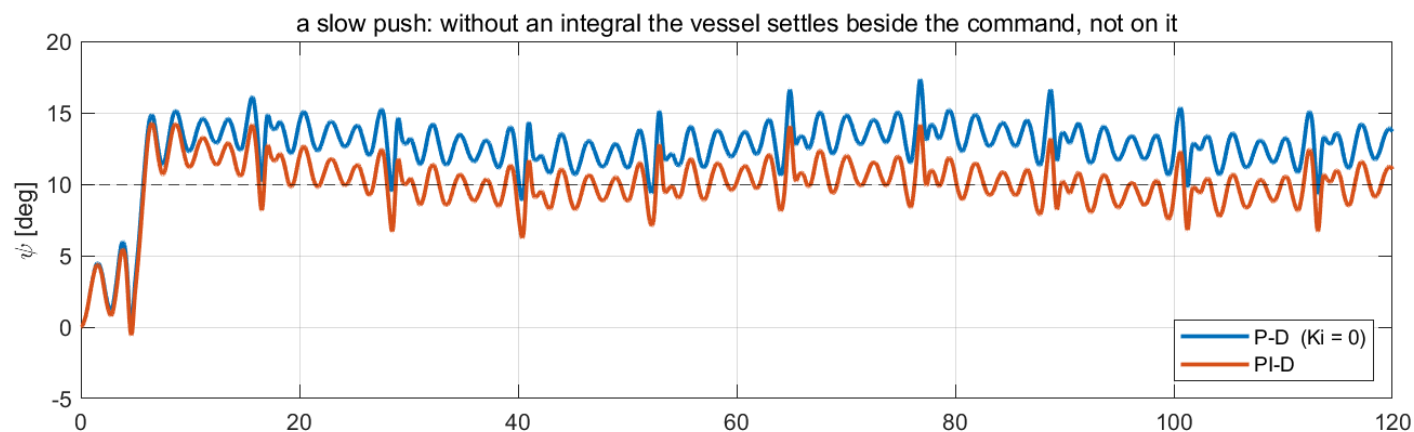
Predict before running. A slow yaw moment of **15 N·m** is switched on. Work out, from $K_p = 300$, the heading error a P-D loop must hold in order to produce a counter-moment of that size — and then say why the integral does **not** wind up on the wave, which is far larger.

Verify. `w07_check(3)`. The window is $t > 60$ s, because the disturbance varies over 60 s and a mean means nothing before one period has passed.

What the checker expects	P-D ($K_i = 0$)	PI-D
mean heading error	3.071°	0.084°
mean moment	-15.13 N·m	-15.13 N·m
heading std	the same on both, to 0.06°	

That last row is the one worth reading twice: the integral removes the offset and **touches nothing else**.

What a correct model produces



Reading the figure	
top, the P-D trace	riding 3° above the dashed command
top, the PI-D trace	on the command, with the same wave ripple
top, the ripple on both	the same size, 1.19° : the integral does not touch the zero-mean part
bottom, both traces	around -15 N·m, following the slow variation
the two mean moments	identical to two decimals — the physics decides the moment, the controller decides who pays for it
the check	if the integral drifts away, the notch is filtering the slow push as well: check that H is unity two decades below ω_0

The point. The notch kept what the controller must act on and removed what it must not. **Filter what cannot be followed; integrate what must be opposed.** A filter that also removed the slow part would be the worst failure of this week — the vessel would be pushed off heading and the measurement would not show it.

Marking

	Weight	What is being marked
Problem 1	40	<code>w07_check(1)</code> passes; the rate channel is filtered too, and the reason is stated
Problem 2	25	<code>w07_check(2)</code> passes; the prediction about phase was written down before running
Problem 3	35	<code>w07_check(3)</code> passes; the reason the wave does not wind up the integral is argued rather than quoted

If something goes wrong

Symptom	Cause	Fix
The model has no To Workspace block whose variable name is <code>flog</code>	only the provided <code>xlog</code> is there	add the second log: three signals, <code>Structure With Time</code>
the notch removes nothing – is it wired?	the filtered signals were built but the autopilot still reads ψ_m	wire ψ_f and r_f into the autopilot
The heading std barely moves	the notch is on the heading only	filter the rate as well: it carries 11.64 deg/s
<code>zeta_n = zeta_d</code> leaves it untouched fails	the filter is not a pure $H(s)$ — an extra lag or gain has crept in	numerator and denominator must differ in one coefficient only
The vessel drifts off heading with a large steady error	the notch is attenuating the slow disturbance too	ω_0 is wrong: check $2\pi/T_0$
Problem 2's two runs are identical	<code>wave_on = 0</code> was set but ζ_n was not restored	the checker sets both; do not set them in the model
The integral runs away	no back-calculation, or K_b has the wrong sign	$I += h(K_i e + K_b(N - u))$, with $K_b > 0$

Reference answers are in `../solutions/`. Read them **after** attempting the problem.